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Part I
The Rising Sea

1 1.2.S.

Tag 01KR, which points to this MathOverflow post.

2 1.2.10.

Proposition 1. *In the category of divisible groups, the map $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ is a monomorphism.*

For divisible groups, here we may consider it as abelian groups that are divisible. Thus we've obtained a full subcategory of an Abelian category. We thus consider ... One important fact is to realise the subgroup of $\mathbb{Z}$ that is divisible is 0.

3 1.3.F. EXERCISE.

A post discussing this problem is HERE.

4 1.3.N

A crutial step is to define the map such that the diagram commute. In order to prove it satisfies the universal property. We're given A be arbitrary with map $g : A \to X$ and $f : A \to Y$.

$$\begin{tikzcd}
A \arrow[dr, dashed, "\exists!\varphi"] \arrow[drr, bend left, "f"] \arrow[ddr, bend right, "g"'] & & \\
& X \times_Z Y \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & Y \arrow[d, "\beta"] \\
& X \arrow[r, "\alpha"'] & Z
\end{tikzcd}$$

We can to define

$$\begin{aligned} \varphi : A &\to X \times_Z Y \quad \text{by} \\ a &\mapsto (g(a), f(a)). \end{aligned}$$

And we can verify this definition will make the diagram commute, and is unique.

5 1.3.O

It's indeed intersection. A post HERE.
A post HERE.

6 1.3.P.

Say we have $X \times Y$ and $X \times_Z Y$. By universal property of product and fibered product we can produce two unique map goes in between. Their composition must be identity, hence they're isomorphic. Notice it's important for Z being a final object.

$$\begin{tikzcd} X \times Y \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & Y \arrow[d, "\beta"] \\ X \arrow[r, "\alpha"'] & Z \end{tikzcd}$$

When we trying to make a map from $X \times Y$ to $X \times_Z Y$, we have to make sure two maps

$$\beta \circ \pi_1 = \alpha \circ \pi_1$$

on $X \times Y$. While Z is the final object, hence they must agree.

There's a cleaner way to state it HERE. Crutial part is applying final property of object Z.

7 1.3.Q.

$$\begin{tikzcd} A \arrow[dr, dashed, "\exists!\varphi"] \arrow[drr, bend left, "g_1"] \arrow[ddr, bend right, "g_2"'] \arrow[ddr, dashed, "\exists!\psi"'] & & \\ & U \arrow[r, "f_1"] \arrow[d, "f_2"] & V \arrow[d, "f_3"] \\ & W \arrow[r, "f_4"'] \arrow[d, "f_5"] & X \arrow[d, "f_6"] \\ & Y \arrow[r, "f_7"'] & Z \end{tikzcd}$$

Label the maps as indicated. To prove the universal property with respect to the "outside rectangle", we're given

$$f_6 f_3 g_1 = f_7 g_2$$

agree on A. While W is fibered product, apply universal property of fibered product with resepct to W we immediately get a unique map

$$\psi : A \to W$$

that makes the diagram involving A, W, X, Y, Z commute. In particularly, we know $f_4\psi = f_3 g_1$. Furthermore, recall that U is the fibered product. We're given the condition that $f_4\psi = f_3 g_1$, by universal property of U we know there exists a unique map

$$\varphi : A \to U$$

making the diagram involving A, U, V, W, X commute. And we claim that the diagram involving A, U, V, Y, Z commute. This is because

$$g_2 = f_5\psi = f_5 f_2 \varphi = (f_5 f_2)\varphi \quad \text{and} \quad g_1 = f_1\varphi.$$

And this proves that U is the fibered product for the diagram involving A, U, V, Y, Z.

A post is HERE.

8 1.3.R

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[rrd, bend left, "\pi_1"] \arrow[rdd, bend right, "\pi_4"'] & & \\
& X_1 \times_Z X_2 \arrow[r, "\pi_2"] \arrow[d, "\pi_3"'] & X_2 \arrow[d, "f_3"] \\
& X_1 \arrow[r, "f_2"'] & Y \arrow[dr, "f_1"] \\
& & & Z
\end{tikzcd}$$

By the universal property of $X_1 \times_Z X_2$, we know there exists a unique map

$$\varphi : X_1 \times_Y X_2 \to X_1 \times_Z X_2$$

"Natural morphism", a convention discussed HERE.

9 Course Notes from Cornell

See HERE.

10 1.3.S. Magic Diagram

Didn't finish. Need to See HERE, HERE!!!

11 1.3.Y. (a)

Yoneda's Lemma Given what we have, define $g : A \to A'$ as

$$g := i_A(\mathrm{id}_A).$$

This is correct, see a post HERE.

12 1.4.C.

(a) See "A Term of Commutative Algebra", Example 7.3 on Page 52.

13 1.6.B.

Write out everything by definition, and we can finish the proof immediately by applying rank-nullity theorem for linear transformation...

14 2.2.6. Definition: Sheaf.

Comments on $\mathscr{F}(\emptyset)$. In category **Set**, the empty set is initial object and one element set is terminal. See Wiki's examples HERE.

14.1 Example

15 2.2.B. Presheaves that are not SHEAVES.

15.1 (a)

For (a): see Wiki's counterexample HERE, which gave an explanation for presheaves on $\mathbb{R}$ instead of $\mathbb{C}$. See a post HERE.

15.2 (b)

See a post HERE, HERE, and HERE.

Proof.
This problem is based on some knowledge from complex analysis. Here are some facts:

- When does a complex function have a square root?
- Theorem 6.2 on Page 100 of [**?**]. This means $f(x) = x$ as a function on $\mathbb{C}$ does not admit a square root for it will vanish.

However, we can cover $\mathbb{C}$ by two slit regions $U_1 = \mathbb{C} - (-\infty, 0], U_2 = \mathbb{C} - [0, \infty)$. And on each U_i, $f\mid_{U_i}$ admits a square root and satisfy gluability axiom.

The solution given in HERE is saying on annulus, $f(x) = x$ cannot have a square root. And then we can follow argument of HERE.

□

16 2.2.7.

See Daping's Notes Definition 2.5 on Page 4 HERE; also a post HERE; also a post HERE.

See a post HERE and HERE regarding gluing sheaves.

17 2.2.C.

Since $\cup U_i$ is colimit for $\{U_i\}$, then $\mathscr{F}(\cup U_i)$ will be limit because of the contravariance. ?

See an expository post HERE.

See another post, with more detailed explanations HERE. "In fancy language, it's stack"...

18 2.2.D.

(b) Motivating example for definition of sheaf.

19 2.2.E.

See comments for Exercise 1.1. of Chapter 2 [**?**].

20 2.2.F.

Almost by definition.

21 2.2.G.

21.1 (a)

22 2.2.9.

See a post HERE.

23 2.2.10.

23.1 References

See this regarding constant presheaf. It's different from a post HERE, and Wiki's page on Constant pre-Sheaf. Why???

23.2

Clearly it's a contravariant functor

$$\mathscr{F} := \underline{S}_{\text{pre}} : \mathbf{Top}(X) \to \mathbf{Set}$$

Let $X = \{a, b\}$ with discrete topology. Pick two sections

$$s_1 \in \mathscr{F}(\{a\}) = S,\ s_2 \in \mathscr{F}(\{b\}) = S$$

such that $s_1 \neq s_2$ given S has at least two distinct elements. Clearly we have

$$s_1 \mid_{\{a\}\cap\{b\}} = s_1 \mid_{\emptyset} = e = \cdots = s_2 \mid_{\{a\}\cap\{b\}} .$$

If it's a sheaf then there exists a global section $s \in \mathscr{F}(\{a, b\}) = S$ such that

$$s_1 = s \mid_{\{a\}} = s \mid_{\{b\}} = s_2,$$

contradiction. It follows that constant presheaf defined this way is not necessarily a sheaf.

24 2.2.E.

We have to deceptively identical pre-sheaves $\mathscr{F}_1$ defined as locally closed, and $\mathscr{F}_2$ defined by giving S discrete topology...

We wish to prove they, as pre-sheaves, are isomorphic. Equivalent, we need to exhibit a natural transformation that admits an inverse. And it suffices to prove by element inclusions:

- Let $f : U \to S$ be a map that's locally constant. Now we take $g(u) = f(u)$ as a map $g : U \to S$ with S endowed with a discrete topology. We claim that g is continuous. It suffices to check for each $s \in S$, the fiber $g^{-1}(s)$ is open. For any point $a \in g^{-1}(s) = f^{-1}(s)$, there exists an open neighborhood $V_a \subset f^{-1}(s)$ such that

  $$f(V_a) = \{s\}$$

  given f is locally constant. While $V_a \subset g^{-1}(s)$, therefore we know $g^{-1}(s)$ is open and g is continuous.

- Conversely, we assume $g : U \to S$ with S given a discrete topology is continuous. We claim $f = g$ is locally constant. For any point $p \in U$, there is an open neighborhood

  $$g^{-1}(f(p)) \ni p$$

  such that f is constant because $f(g^{-1}(f(p))) = \{p\}$.

Now we try to check constant sheaf $\mathscr{F}$ is indeed a sheaf. We're going to prove identity axiom and gluability axiom using the "better description", which is much easier to check:

- If we have two functions, which equal whenever we restric to any open subset from an open cover, then they must be equal. For functions are precisely defined this way.

- Define the global section for any choice, and it's going to be well-defined for they're compatible.

Therefore $\mathscr{F} = \underline{S}$ is indeed a sheaf.

25 2.2.F.

Same argument as 18.

26 2.2.G.

26.1 (a)

It's clearly a pre-sheaf.

Fix an open subset $U \subset X$ with an open cover $\{U_i\}_{i \in I}$ for some index set I. Denote the presheaf as $\mathscr{F}$.

Pick two continuous maps $s_1, s_2 : Y \to X$ that satisfying the requirements, i.e. $s_1, s_2 \in \mathscr{F}(U)$.

Both functions will agree on U since

$$\mathrm{Res}_{U,U_i} \ \ s_1 = \mathrm{Res}_{U,U_i} \ \ s_1$$

for arbitrary U_i, whose union is U. So we must have $s_1 = s_2$.

Again with this open cover $\{U_i\}_{i \in I}$ and $a_i \in \mathscr{F}(U_i)$ for $i \in I$. Equivalently, we know $a_i : U_i \to Y$ is a continuous map satisfying $\mu \circ a_i = \mathrm{Id}\,|_{U_i}$. Now let's define a map

$$\begin{aligned} f : U &\to Y \\ u &\mapsto a_i(u) \ \ \text{when} \ \ u \in U_i. \end{aligned}$$

It's well-defined by our assumption. Also it's continuous since preimage of an open set in $V \subset Y$ is a union of open subsets given by continuity of each a_i. Similarly we can check $\mu \circ f = \mathrm{Id}\,|_U$ as expected.

Unverified ?

26.2 (b)

27 2.2.11. Espace Étalé

See a post discussion accent letter in LaTeX HERE.

See an exercise in [**?**] Chapter 2, Exercise 1.13.

See the discussion after Lemma 7. on Page 229 of [**?**].

For *section*, see Wiki's explanation for *section* in context of fiber bundle; and *section* in terms of category theory.

See a detailed post HERE.

28 2.2.H.

Clearly it's again a contravariant functor, therefore $\pi_* \mathscr{F}$ must be a pre-sheaf. When $\mathscr{F}$ is a sheaf, I checked identity axiom (lots of things to write down).

29 2.3.A.

I'm planning to use universal property to define the induced map ϕ_P.

One crutial step is to verify the diagram below is commutative

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r, "\rho_{UV}"] \arrow[d, "\phi(U)"'] & \mathscr{F}(V) \arrow[d, "\phi(V)"] \\
\mathscr{G}(U) \arrow[r, "\tau_{UV}"'] \arrow[dr] & \mathscr{G}(V) \arrow[d] \\
& \mathscr{G}_P
\end{tikzcd}$$

And this is because the square diagram in the upper half commute given ϕ is a natural transformation; the lower half is by definition of $\mathscr{G}_P$. Then by universal property of colimit induces a map

$$\phi_P : \mathscr{F}_P \to \mathscr{G}_P$$

which makes the diagram commute.

See a post defined the map HERE.

30 2.3.B.

To define a functor $\pi_* : \mathbf{Set}_X \to \mathbf{Set}_Y$. Firstly, we have to define for any $\mathscr{F} \in \mathbf{Set}_X$,

$$\pi_*(\mathscr{F})(U) = \mathscr{F}(\pi^{-1}(U))$$

for any $U \in \mathfrak{Top}(X)$ as in **??**.

Secondly, for any natural transformation $\phi : \mathscr{F} \to \mathscr{G}$, we define $\pi_*(\phi)$ by specifying

$$\pi_*(\phi)(U) \ \mapsto \mathscr{F}(\pi^{-1}(U)) \to \mathscr{G}(\pi^{-1}(U)).$$

? Is this correct

31 2.3.C.

This is Exercise 1.15. from Chapter II of [**?**] on Page 67.

Proof.
Clearly $\mathrm{Hom}(\mathscr{F}, \mathscr{G})(U)$ takes value in the set of all natural transformations from $\mathscr{F}\mid_U$ to $\mathscr{G}\mid_U$. Namely, we have

$$U \mapsto \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U).$$

The restriction map induced by $V \subset U$ is given by consider a natural morphism $\alpha \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ as a natural morphism from

$$\mathscr{F}\mid_V \to \mathscr{G}\mid_V .$$

So set-theoretically restriction map is identity map, with exception that it regard an element as a presheaf on a smaller open subset.

?

Hence $\mathrm{Hom}(\mathscr{F},\mathscr{G})(\cdot)$ is a presheaf.

Fix an open subset $U \subset X$, with an open covering $\{U_i\}_{i\in I}$ for some index set I. Pick two natural transformations $\alpha, \beta \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$. Assume for any $i \in I$,

$$\mathrm{Res}_{U,U_i}\,\alpha = \mathrm{Res}_{U,U_i}\,\beta.$$

More precisely, this means for any open subset $V_i \subset U_i$ where $i \in I$ is arbitrary, we have

$$\alpha\mid_{U_i}(V_i) = \beta\mid_{U_i}(V_i).$$

However, note $\{U_i\}_{i\in I}$ is an open cover for the whole space U we're considering. It follows that for any open subset $W \subset U$, we can denote $W_i = W \cap U_i$ and express W as a union of W_i where $i \in I$.

$$\begin{aligned}
\alpha(W) =& \alpha\left(\bigcup_{i\in I} W_i\right) \in \mathrm{Obj}(\mathbf{Set}) \\
=& \bigcup_{i\in I} \alpha(W_i) \\
=& \bigcup_{i\in I} \alpha\mid_{U_i}(W_i) \\
=& \bigcup_{i\in I} \beta\mid_{U_i}(W_i) \\
=& \cdots \\
=& \beta(W).
\end{aligned}$$

Here the third equality holds by the definition of restriction map. While $W \subset U$ is arbitrary, it follows that $\alpha = \beta \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ as expected.

It remains to check gluability. Again $\{U_i\}_{i\in I}$ is an open cover of U. Pick natural transformations $\alpha_i \in \mathrm{Mor}(\mathscr{F}\mid_{U_i}, \mathscr{G}\mid_{U_i})$. They're compatiable in the sense that for any open subset $W_{ij} \subset U_i \cap U_j$, we know

$$\mathrm{Res}_{U_i,U_i\cap U_j}\,\alpha_i = \mathrm{Res}_{U_j,U_i\cap U_j}\,\alpha_j \;\Rightarrow\; \alpha_i(W_{ij}) = \alpha_j(W_{ij}).$$

Now we try to define a natrual transformation $\alpha \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)$ such that $\alpha\mid_{U_i} = \alpha_i$. For any open subset $Y \subset U$, denotes $Y_i := Y \cap U_i$.

$$\begin{aligned}
\alpha(Y) : \mathscr{F}\mid_U(Y) &\to \mathscr{G}\mid_U(Y) \\
\mathscr{F}\mid_U(Y_i) \ni x &\mapsto \alpha_i(x).
\end{aligned}$$

This is map in sets, it's well-defined for $\{Y_i\}_{i\in I}$ is an open covering for Y and each α_i is compatiable. By construction we know $\alpha\mid_{U_i} = \alpha_i$. Hence we've checked gluability.

Need to check, but I think it's basically unwrapping a long long defintion

$\square$

31.1 Verification

See a post HERE.

Also see a lemma from Stacks Project HERE. This lemma basically proves gluability and uniqueness, based on the fact that Sheaf Hom is already a pre-sheaf. In the proof of the above lemma, we defined the natural transformation in the way such that the following diagram commute

$$\begin{tikzcd} s \in \mathscr{F}(U) \arrow[r, "\varphi_U"] \arrow[d] & \mathscr{G}(U) \arrow[d] \\ \mathscr{F}(V) \arrow[r, "\varphi_V"'] & \mathscr{G}(V) \end{tikzcd}$$

And this relies on the fact that $\mathscr{G}$ is a sheaf!, by looking at every $U \cap U_i$, which covers U.

31.2 Warning

Sheaf Hom does not commute with taking stalks. But there exists at least one map from

$$\begin{aligned} &\mathrm{Hom}(\mathscr{F}, \mathscr{G})_p \to \mathrm{Hom}(\mathscr{F}_p, \mathscr{G}_p) \\ &\{(\alpha, U) \ \mid \ p \in U, \ \alpha \in \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U)\}/\sim_1 \mapsto ... \end{aligned}$$

31.3 Warning: References

See a post HERE, which contains a link to the detailed version of the counterexample HERE. According to post HERE, The direction was correct.

See Stefan's notes on Page 18 for a concrete example HERE. One comment on "Hom functor preserve limit" HERE.

Also, the first argument $\mathscr{F}$ doesn't have to be a sheaf, a pre-sheaf is enough. See a post HERE.

31.4 Counterexample

I'll add some details to the first example mentioned in the post linked above.

Proof.
For any $U \subset X$, where $\mathscr{F}$ is skycraper sheaf at $p \in X$ with value group A and $\mathscr{G}$ is a constant sheaf on topological space X with value group A. We claim that

$$\mathcal{H}om(\mathscr{F}, \mathscr{G})(U) = \mathrm{Mor}(\mathscr{F}\mid_U, \mathscr{G}\mid_U) = 0$$

as an Abelian group for arbitrary $U \subset X$.

It suffices to check the above statment is correct when $U = X$ for $\mathcal{H}om(\mathscr{F}, \mathscr{G})$ is a sheaf, i.e. we need to show the group

$$\mathcal{H}om(\mathscr{F}, \mathscr{G})(X) = \mathrm{Mor}(\mathscr{F}, \mathscr{G}) = 0$$

Pick any natural transformation $\alpha \in \mathrm{Mor}(\mathscr{F}, \mathscr{G})$. We wish to show the group homomorphism $\alpha(U)$ is 0 (i.e. sends everything to 0 in the codomain $\mathscr{G}(U)$). For an open subset U, it admits an open covering $\{U_i\}$ in which every U_i is connected. While $\mathscr{G}$ is a sheaf, to prove $\alpha(U) = 0$ it suffices to check $\alpha(U_i) = 0$. Fix i and denote $U_i = U_0$. When $p \notin U_0$, then $\mathscr{F}(U_0) = 0$ and the map $\alpha(U_0) = 0$.

Now assume $p \notin U_0$. We can still argue $\alpha(U_0)$ is 0 group homomorphism by restrict it to a smaller open subset that doesn't contain p, because the skycraper sheaf will be 0 group.

Assume p is closed an not open, which means it's not isolated. Then $V := U \setminus \{p\}$ is an open subset that doesn't contain p, hence $\mathscr{F}(V) = 0$ by definition. ?

$$\begin{tikzcd}
\mathscr{F}(U_0) \arrow[r, "\alpha(U)"] \arrow[d, "\mathrm{Res}_{U_0 V}"'] & \mathscr{G}(U_0) = A \arrow[d] \\
\mathscr{F}(V) = 0 \arrow[r] & \mathscr{G}(V) = \oplus A
\end{tikzcd}$$

Notice that we assumed U_0 to connected, therefore $\mathscr{G}(U_0) = A$ is composed of single one copy of A. While $\mathscr{G}(V) = \oplus_{j \in J} A$ for some index set J. We claim that $\mathscr{G}(U_0) \to \mathscr{G}(V)$ is an injection:

- We claim that the map is in fact diagonal map by considering the following commutative diagram

$$\begin{tikzcd}
a \in \mathscr{G}(U_0) \arrow[r] \arrow[d, hook] & (a, a, ..., a)_{j \in J} \in \mathscr{G}(V) \arrow[d, hook] \\
\amalg_{q \in U_0} \mathscr{G}_q = \amalg_{q \in U_0} A & \amalg_{q \in V} A \arrow[l, hook']
\end{tikzcd}$$

  Because the map from $\mathscr{G}(U_0) \to \amalg_{q \in U_0} \mathscr{G}_q = \amalg_{q \in U_0} A$ is the diagonal map, therefore the map we considering must factor through it. Hence it could only be the diagonal map, with index set J.

Therefore by commutativity of the diagram together with the injectivity of the above map we can conclude $\alpha(U_0) = 0$ as expected.

$\square$

31.5 Abelian group structure

The Abelian group structure on $\mathcal{H}om(\mathscr{F}, \mathscr{G})(U)$ is given by defining

$$\alpha + \beta(U) : \mathscr{F}(U) \to \mathscr{G}(U) \ x \mapsto \alpha(U)x + \beta(U)x$$

for two natrual transformations $\alpha, \beta \in \mathcal{H}om(\mathscr{F}, \mathscr{G})(U)$. Notice that $\alpha + \beta$ is indeed a natural transformation, because it's compatible with restriction maps, which we could check by definition... Element $0 \in \mathcal{H}om(\mathscr{F}, \mathscr{G})(U)$ is just a natural transformation sends everything to $0 \in \mathscr{G}(U)$.

32 2.3.D.

32.1 (a)

Since the stalk doesn't commute with sheaf hom, I guess it could be more complicated to compute the stalk for sheaf hom $\mathcal{H}om$. So I'll check two functors are isomorphic in terms of *natural transformation*.

Let $\mathscr{C} := \underline{\{p\}}$ denotes the constant sheaf with value set $\{p\}$. Firstly, by definition we know for any open subset $U \subset X$

$$\mathcal{H}om(\mathscr{C}, \mathscr{F})(U) = \text{ Mor}(\mathscr{C}|_U, \mathscr{F}|_U).$$

Pick any element $\alpha \in \text{Mor}(\mathscr{C}|_U, \mathscr{F}|_U)$, which is a natural transformation.

Endow $\{p\}$ with discrete topology. For any open subset $V_1 \subset U$, we have set map $\alpha(V_1)$

$$\alpha(V_1) : \ \mathscr{C}|_U(V_1) = \mathscr{C}(V_1) \longrightarrow \mathscr{F}|_U(V_1) = \mathscr{F}(V_1).$$

The domain only contains one element: the set is all continuous map from some open subset in V_1 to $\{p\}$, hence there's only such function and we denote it as f. It follows that the map is completely determined by where f goes, $\alpha(V_1)$ could be identified with an element in $\mathscr{F}(V_1)$. In particular, for any $\alpha \in \text{Mor}(\mathscr{C}|_U, \mathscr{F}|_U)$, $\alpha \in \mathscr{F}(U)$, and

$$\mathcal{H}om(\mathscr{C}, \mathscr{F})(U) = \text{Mor}(\mathscr{C}|_U, \mathscr{F}|_U) = \mathscr{F}(U)$$

for any $U \subset X$, notice that it's in fact set-theorically bijection, hence it's a natural isomorphism and therefore two functors are isomorphic as expected.

Need to verify

32.2 References

A potentially related post HERE.

33 2.3.E.

- Diagram Chase: ?
- Universal Property: According to Definition 1.6.4., Kernel was defined using universal property. Therefore we can use universal property of $\text{Ker}(\phi(U))$ to deduce the unique existence of such map for $\text{Ker}(\phi(V)) \to \mathscr{G}(U) = 0$.

34 2.3.G. 2.3.H.

Notice that we're talking about pre-sheaves here. Hence these properties are satisfied by definition...

35 2.3.J.

Question: wouldn't it be better if $\underline{\mathbb{Z}}$ is replaced by $\underline{\mathbb{C}}$?

35.1 References

See a post HERE. In fact, the quotient sheaf could be not separated, see a post HERE.

See a detailed lecture note HERE.

And a solution HERE and HERE.

Here's another example where image presheaf might not be a sheaf.

36 2.4.A.

36.1 Hint

See G ortz's book [**?**], page 52, Prop 2.23. (1); or Lemma 2.9. on Page 35 of [**?**].

37 2.4.2. Compatible Germs

See a post regarding compatible germs and Espace Étalé.

37.1 Two Equivalent Definitions

The following two definitions are equivalent:

- *Element* $(s_p)_{p\in U} \in \prod_{p\in U} \mathscr{F}_p$ *consists of compatible germs* ...
- *There is an open covering* $\{U_i\}$ *of* U, ...

Proof.
For any $p \in U$, pick some representative of $s_p \in \mathscr{F}_p$ as follow:

$$(U_p, \tilde{s}_p)$$

where open subset $U_p \subset U$ contains p with $\tilde{s}_p \in \mathscr{F}(U_p)$. Then we take

$$\{U_p\}_{p\in U}$$

as an open covering of U and take corresponding sections $\tilde{s}_p \in \mathscr{F}(U_p)$ for $p \in U$. We can check conditions for the second statement is satisfied: Clearly $p \in U_p$ and s_p is the germ of $\tilde{s}_p$ at p for the first condition required this is true for any $q \in U_p$.

Conversely, pick any $p \in U$. We can find some U_i that contains p with a section $f_i \in \mathscr{F}(U_i)$ such that s_p is the germ of f_i at p. So we take

$$(f_i \in \mathscr{F}(U_i),\ U_p) \in \mathscr{F}_p$$

as representative of f_i. Notice that germ of f_i at any $q \in U_i$ is s_q as expected. $\square$

37.2 Section induces a choice of compatible germs

Any sections s of $\mathscr{F}$ over U gives a choice of compatible germs for U.

Proof.
Let $s \in \mathscr{F}(U)$ as given and pick any open covering $\{U_i\}_{i\in I}$ of U where I is an index set. And define $f_i := s|_{U_i} \in \mathscr{F}(U_i)$. Now we may assume $p \in U_i$ for some U_i. The following diagram is commute for any $p \in U_i$, with element being the image under corresponding maps.

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r] & \mathscr{F}(U_i) \arrow[r] & \mathscr{F}_p \\
s \arrow[r, mapsto] \arrow[rr, bend right, mapsto] & f_i := s|_{U_i} & s_p
\end{tikzcd}$$

Assume $p \in U_i$ for a fixed $i \in I$, then

$$s_p = (\mathscr{F}(U) \to \mathscr{F}_p)(s) = (\mathscr{F}(U_p) \to \mathscr{F}_p)(f_i)$$

by commutativity of the above diagram. Hence s_p is germ of f_i at p as expected. $\square$

38 2.4.B.

$$\mathscr{F}(U) \to \prod_{p\in U} \mathscr{F}_p$$

In the last subsection of Section 37, we checked that any section will induce something called compatible germs on the left of the above map. Question is that what are the images of the above map? The following lemma will give us complete description of the image of the map above.

38.1

Any choice of compatible germs is the image of a section.

Proof.
I'll take the second definitions of a choice of compatible germs: there is an open cover $\{U_i\}_{i\in I}$ of U where I is understood as an index set, and there are sections $f_i \in \mathscr{F}(U_i)$ such that s_p is the germ of f_i at any $p \in U_i$.

Let's consider two open subsets $U_i, U_j \subset U$ where $i, j \in I$ with $f_i \in \mathscr{F}(U_i), f_j \in \mathscr{F}(U_j)$. Assume $U_i \cap U_j \neq \emptyset$. Then for any $p \in U_i \cap U_j$, s_p is the germ of f_i and f_j at p, i.e.

$$[U_i, f_i] = s_p = [U_j, f_j] \in \mathscr{F}_p.$$

Hence there exists open subset $W_{ij,p} \subset U_i \cap U_j$ containing p such that $f_i|_{W_{ij,p}} = f_j|_{W_{ij}}$. Notice that $p \in U_i \cap U_j$ is arbitrary, therefore for any p there exists some $W_{ij,p}$ (depending on p). And all such $\{W_{ij,p}\}_{p \in U_i \cap U_j}$ will be an open covering for $U_i \cap U_j$, by Identity Axiom we know

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}.$$

While i, j were arbitrary, Gluability Axiom implies there exists $f \in \mathscr{F}(U)$ such that $f|_{U_i} = f_i$. And we can check that f is indeed the preimage of the given choice of compatible germs: pick some U_i containing p,

$$f_p = (\mathscr{F}(U_i) \to \mathscr{F}_p)(f|_{U_i})$$

Hence f_p is the germ of f_i at p.

□

38.2 Hint

See this post.

39 2.4.C.

According to the hint, we have the following diagram commute for according to construction of 2.3.A.

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r, shift left, "\phi_1(U)"] \arrow[r, shift right, "\phi_2(U)"'] \arrow[d, "\pi"'] & \mathscr{G}(U) \arrow[d, hook, "\iota"] \\
\prod_{p \in U} \mathscr{F}_p \arrow[r, "\Phi"'] & \prod_{p \in U} \mathscr{G}_p
\end{tikzcd}$$

Hence we have

$$\iota \circ \phi_1(U) = \Phi \circ \pi = \iota \circ \phi_2(U) \;\Rightarrow\; \phi_1(U) = \phi_2(U)$$

for ι is injective.

40 2.4.D.

See Hartshorne [?] Chapter II, Proposition 1.1.

40.1 Warning

It could happen that two non-isomorphic sheaves have isomorphic stalks. See Sheaf Hom Section 31 regarding skycraper sheaf and constant sheaf.

See a post HERE.

41 2.4.E.

41.1 (a)

Pick a non-separated presheaf: See Stacks Project Definition 007A ?
Here is an example of gluable unseparated presheaf.
See a post discussing two examples where section isn't determined by stalks.

41.2 (b)

For Section 39, it's important to assume $\mathscr{G}$ is a sheaf.

See the answer by Andreas Blass HERE. For two pre-sheaves $\mathscr{F}, \mathscr{G}$, we might encounter a counterexample as follows:

Let $X = \{p, q\}$ be a discrete topological space with two points. Define two pre-sheaves $\mathscr{F}, \mathscr{G}$ on X with values in set $\{0, 1\}$. We specify two pre-sheaves as follow:

$U \subset X$	$\{p, q\}$	$\{p\}$	$\{q\}$	$\emptyset$
$\mathscr{F}(U)$	$\{0, 1\}$	$\{0\}$	$\{0\}$	$\{0\}$
$\mathscr{G}(U)$	$\{0, 1\}$	$\{1\}$	$\{1\}$	$\{1\}$

where the restriction maps for $\mathscr{F}$ and $\mathscr{G}$ are the only possible ones. Now we wish to define two different morphism of sheaves $\alpha, \beta : \mathscr{F} \to \mathscr{G}$ as natural transformations. For all open subsets except $U = \{p, q\}$, we have

$$\alpha(U) = \beta(U) : \{0\} \to \{1\}.$$

The place we intentionally make them different is when $U = \{p, q\}$:

$$\begin{aligned} \alpha(U) &: \{0, 1\} \to \{0, 1\} \quad \text{by} \quad 0 \mapsto 0,\ 1 \mapsto 1; \\ \beta(U) &: \{0, 1\} \to \{0, 1\} \quad \text{by} \quad 0 \mapsto 1,\ 1 \mapsto 0. \end{aligned}$$

Clearly as morphisms $\alpha \neq \beta$. But we have

$$\begin{aligned} \alpha_p \times \alpha_q : \prod_{x \in \{p,q\}} \mathscr{F}_x &\to \prod_{x \in \{p,q\}} \mathscr{G}_x \\ \{0\} \times \{0\} &\mapsto \{1\} \times \{1\}. \end{aligned}$$

But notice that there's only **one** choice of induced morphism between $\mathscr{F}_x \to \mathscr{G}_x$ for any $x \in X = \{p, q\}$. This means, even if we're given another different maps β, it will induce the same map on stalks. ?

41.3 (c)

Take a pre-sheaf $\mathscr{F}$ (on space X) that is not a sheaf. For example, constant pre-sheaf. We know the stalk of its sheafification at any $p \in X$ is

$$\mathscr{F}_p = \mathscr{F}_p^\sharp.$$

42 2.4.O.

A post HERE, HERE.

Another related post explaining isomorphic sheaves will give rise to injective and surjective maps on sections, this does not contradict the fact that surjectivity of morphism of sheaves cannot have analogus statement on sections.

See Richard on YouTube, Scheme Lecture 03 last example. Let $U = \mathbb{C} \setminus \{0\}$ with classical topology.

$$\mathscr{O}_X(U) \to \mathscr{O}_X^*(U)$$

is not surjective for $x \in \mathscr{O}_X^*(U)$ doesn't admits a holomorphic log for $f(0) = 0$ while exponential function can never be 0. Because we cannot find $logx$ as a holomorphic function on $\mathbb{C}$.

See a very detailed solution HERE.

42.1 Complex analysis background

See Theorem 6.1., 6.2. Chap 3 Page 98 [**?**].

See this great POST!

See a note HERE. A post HERE.

43 2.5.1.

43.1 Remarks

We can also define, for general open subset by taking limit: see comments of 009H by [**?**]; and [**?**] Proposition 2.20. on Page 51 with discussion above it.

44 2.5.B.

Proof.

... □

44.1 Another Proof

Proof.

A potentially wrong approach is to recognise that, in the definition of $\mathscr{F}(U)$: we set ?

$$\mathscr{F}(B) := \{(f_p \in F_p)_{p \in B} \ \mid \ \star\}.$$

The condition $\star$ is exactly the definition of compatible germs. While $F(B)$ could be identified with it's image in an injective map

$$F(B) \to \prod_{p \in B} F_p.$$

We proved in Section 38 that the image is precisely the compatible germs, therefore we must have they're isomorphic. □

44.2 Remarks

In fact, for sheaf on base F, the map

$$F(U) \to \prod_{p \in U} F_p$$

is injective. The proof is essentially the same as for sheaf on X.

45 2.5.D.

See [**?**] Chap 2 Exercise 2.7. on Page 44. This exercise will be used for Rising Sea [**?**] 4.3.B.

46 2.6.A.

I don't know what "natural" means. Is there a natural transformation?

I'll just check the isomorphism. And assume kernel sheaf has its corresponding the universal property.

Firstly, we have $(\mathrm{Ker}(\mathcal{F} \to \mathcal{G}))_p \to \mathrm{Ker}(\mathcal{F}_p \to \mathcal{G}_p)$ by universal property of $\mathrm{Ker}(\mathcal{F}_p \to \mathcal{G}_p)$. This is because we can compute the composition

$$(\mathrm{Ker}(\mathcal{F} \to \mathcal{G}))_p \to \mathcal{F}_p \to \mathcal{G}_p = 0.$$

For $s_p \in (\mathrm{Ker}(\mathcal{F} \to \mathcal{G}))_p$, we can diagram chasing it...

Conversely, ... a diagram chasing argument.

46.1 Solution

For details see Autumn HW2 2023 Problem 3. Or this online solution.

47 2.7.A.

I couldn't give a complete example ...

47.1 Solution

A post on inverse image presheaf.

Let $X = \{p, q\}, Y = \{w\}$ with discrete topolgoy. Let $\mathscr{G}$ be a constant sheaf on Y with value group A which contains at least two elements. Compute inverse image presheaf.

Proof.
For any $U \subset X$, we have

$$\pi_{\mathrm{pre}}^{-1}\mathscr{F}(U) = \mathrm{Colim}_{V \supset \pi(U) = \{w\}} \mathscr{G}(V) = \mathrm{Colim}_{V = \{w\}} \mathscr{G}(V) = A.$$

Hence it's the inverse image pre-sheaf $\pi_{\text{pre}}^{-1}\mathscr{F}$ is a constant pre-sheaf. It's not a sheaf for exactly the same reason constant pre-sheaf isn't a sheaf for failure of gluability. $\square$?

48 2.7.D.

By definition... See [**?**] Definition on Page 65 of Section 2 of Chapter II.

49 2.7.G.

See **??**.

50 3.1.A.

?

51 3.2.A.

51.1 Example 5

A complete description of $\text{Spec}(\mathbb{R}[x])$ is given by a post HERE.

To find all prime ideals in P.I.D. $\mathbb{R}[x]$ is generated by one single element, i.e. a polynomial $f \in \mathbb{R}[x]$. While $\mathbb{R}[x]$ is U.F.D., a prime element is equivalent to an irreducible element. Therefore we need to find all irreducible polynomials in $\mathbb{R}[x]$. Notice that field extension

$$[\mathbb{C} : \mathbb{R}] = 2 \;\Rightarrow\; \deg f \leq 2$$

for any irreducible polynomial f.

In case of $\deg f = 2$, it's precisely the case where we require f to be an irreducible quadratic.

See a post HERE.

52 3.2.B.

For an irreducible polynomial $x^2 + ax + b \in \mathbb{R}[x]$, it cannot admit a real root. Therefore in algebraic closure of R, i.e. in $\mathbb{C}$, we can find two roots α_1, α_2 of the polynomial.

53 3.2.C.

?

54 3.2.D.

I didn't figure out the Euclid's proof for this.

For Euclid's proof, see a post HERE.

Basic ideal is consider

$$f = \prod_{i \in I} f_i + 1$$

where $\langle f_i \rangle = \mathfrak{p}_i$ for a *finite* index set I. Here $\mathfrak{p}_i \triangleleft k[x]$ is a prime ideal. Each f_i must have degree $\deg f_i \geq 1$ because otherwise $\mathfrak{p}_i$ will contain a unit. While in integral domain $k[x]$, we know degree f is again large or equal to 1, hence not a unit. Clearly it's nonzero. As a non-zero non-unit element in U.F.D. $k[x]$, it could be written as a product of irreducible elements, say g_1. It is will also generate a prime ideal, which means

$$\langle g_1 \rangle = \mathfrak{p}_i = \langle f_i \rangle \;\Rightarrow\; g_1 = u f_i$$

for some $i \in I$ and unit u. Then $f \equiv 1 \pmod{g_i}$, which means g_i will divide unit 1, contradiction.

See a post HERE, HERE.

55 3.2.L. Exercise

Proof.
Localisation commute with quotient, therefore we plan to prove the isomophism by constructing a ring homomorphism from $\psi : \mathbb{C}[x,y]_x \to \mathbb{C}[x]_x$ and compute the kernel. The map ψ is defined as

$$\begin{aligned} \psi : \mathbb{C}[x,y]_x &\to \mathbb{C}[x]_x \\ f(x,y)/x^i &\mapsto f(x,0)/x^i \end{aligned}$$

where $f(x,y) \in \mathbb{C}[x,y]$ and i is some integer.

Clearly we see $(xy)_x \subset \operatorname{Ker}\psi$. Conversely, let's pick an element $f(x,y)/x^i$ such that $f(x,0)/x^i = 0 \in \mathbb{C}[x]_x$, this implies there exists $j \in \mathbb{N}$ such that

$$x^j f(x,0) = 0 \in \mathbb{C}[x] \;\Rightarrow\; f(x,0) = 0 \in \mathbb{C}[x].$$

stopped,...

The conclusion follows immediately if we realise

$$(y)_x = (xy)_x \subset \mathbb{C}[x,y]_x.$$

One way to interpret this is as follow: both $(x)_x$ and $(xy)_x$ are image of localisation map of a principal ideas $(x), (xy) \triangleleft \mathbb{C}[x,y]$. And two principal ideals are the same if they differ by a unit, say x. Or we can perform a double inclusion argument. $\square$

55.1 Hint

See a post HERE.

Some details are not clear, so see the following.

Proof.
Two facts to recall is that we have two exact sequences

$$0 \longrightarrow (xy)_x \subset \mathbb{C}[x,y]_x \longrightarrow \mathbb{C}[x,y]_x \longrightarrow (\mathbb{C}[x,y]/(xy))_x \longrightarrow 0$$

$$0 \longrightarrow (y) \subset \mathbb{C}[x,y] \longrightarrow \mathbb{C}[x,y] \longrightarrow \mathbb{C}[x] \longrightarrow 0$$

Here the second rows will induce, by exactness of localisation

$$0 \longrightarrow (y)_x = (xy)_x \subset \mathbb{C}[x,y]_x \longrightarrow \mathbb{C}[x,y]_x \longrightarrow \mathbb{C}[x]_x \longrightarrow 0$$

And this concludes the desired isomorphism. One crutial observation I missed was $(xy)_x = (y)_x$.

□

56 3.2.T.

I guess it's

?

$$f(x+\epsilon) = f(x) + \epsilon f'(x) \in k[x,\epsilon]/(\epsilon^2)$$

57 3.5.E.

Notice that

$$\begin{aligned} D(f) \subset D(g) \;&\Leftrightarrow\; V(\langle f\rangle) \supset V(\langle g\rangle) \\ \Leftrightarrow\; f \in \sqrt{\langle f\rangle} \subset \sqrt{\langle g\rangle} \;&\Rightarrow\; f^n \in \langle g\rangle \;\text{ for some }\; n \geq 1. \end{aligned}$$

In fact, if we have $f^n \in \langle g\rangle$ for some integer n, then

$$f \in \sqrt{\langle g\rangle} \;\Rightarrow\; \langle f\rangle \subset \sqrt{\langle g\rangle} \;\Rightarrow\; \sqrt{\langle f\rangle} \subset \sqrt{\langle g\rangle}.$$

Hence the equivalence of the first two statements are checked.

Now if we assume $f^n \in \langle g\rangle$ for some integer n, then $cg = f^n$ for some $c \in A$. Hence

$$\frac{g}{1} \cdot \frac{c}{f^n} = 1,$$

which means g is an invertible element of A_f. Conversely, g being an invertible element in A_f means there exists a/f^m such that

$$g/1 \cdot a/f^m = ga/f^m = 1/1 \in A_f \;\Rightarrow\; f^l(ga - f^m) = 0 \in A$$

for some integers l, m. Hence take $n := l + m$ leads to $f^l f^m = f^l ga \in \langle g\rangle$.

57.1 References

Checked at Lemma 01HR of [**?**].

58 3.6.6.

This is related to the counterexample Section 71.

The inclusion map is given by universal property of coproduct: for a fixed $i \in I$, we have a continuous projection $\pi_i : \prod_{j\in\mathbb{N}} A_i \to A_i$ which induces $\operatorname{Spec} A_i \to \operatorname{Spec} \prod_{j\in\mathbb{N}} A_i$. More explicitly, this map on topological spaces is defined by taking pre-image of a prime ideal. Two distinct prime ideals $\mathfrak{p}, \mathfrak{q} \triangleleft A_i$ will give rise to different pre-image of prime ideals. Hence this map is injective. And this induces, by universal property of coproduct i.e. disjoint union of two topological spaces, a map

?

$$\amalg_{j\in\mathbb{N}} \operatorname{Spec} A_i \to \operatorname{Spec} \prod_{j\in\mathbb{N}} A_i.$$

This map is injective for the inclusion from $\operatorname{Spec} A_i \to \amalg_{j\in\mathbb{N}} A_j$ is injective, the commutative diagram proves the injectivity.

$$\begin{tikzcd}
\operatorname{Spec} A_j \arrow[r, hook] \arrow[dr, hook] & \amalg_{j\in\mathbb{N}} \operatorname{Spec} A_j \arrow[d, hook] \\
& \operatorname{Spec} \prod_{j\in\mathbb{N}} A_j
\end{tikzcd}$$

59 4.1.1. Definition: Structure Sheaf on Base

As stated, we define

$$\begin{aligned}
\mathscr{O}_{\operatorname{Spec} A}(D(f)) &:= S_{D(f)}^{-1} A \\
&:= \{g \in A \mid V(g) \subset V(f)\}^{-1} A.
\end{aligned}$$

This is well-define, i.e. independent of the choice f because when we have $D(f') = D(f)$, the localisation will be the same.

Also, we claim $S_{D(f)}$ is indeed a multiplicatively closed subset of A:

- $1 \in S_{D(f)}$ for $V(1) = \emptyset \subset V(f)$ for any choice of $f \in A$.
- Assume $g_1, g_2 \in S$, then

$$D(f) \subset D(g_1 g_2) = D(g_1) \cap D(g_2) \;\Rightarrow\; V(g_1 g_2) \subset V(f).$$

60 Definition of Restriction Map

Here are some remarks for the restriction map. Notice that we have

$$D(f') \subset D(f) \;\Rightarrow\; S^{-1}_{D(f)} \subset S_{D(f')}.$$

Hence the restriction is just inclusion map.

?

60.1 References

The map is given explicitly on Page 4 on the BOOK.

61 4.1.A.

$$\begin{tikzcd}
 & A \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\
A_f \arrow[rr, dashed, leftrightarrow, bend right, "\exists!"'] \arrow[rr, dashed, leftrightarrow, bend left, "\exists!"] & & S^{-1}_{D(f)}A = \mathscr{O}_{\operatorname{Spec} A}(D(f))
\end{tikzcd}$$

Here α, β denotes the localisation map. For β, we have $f \in S_{D(f)}$ for $V(f) \subset V(f)$. Since it's a multiplicatively closed set, then

$$\beta(\{1, f, f^2, ...\}) \subset (S^{-1}_{D(f)}A)^\times$$

will become units, therefore by universal property of A_f there exists a unique map from A_f to $S^{-1}_{D(f)}A$ making the diagram commute.

Conversely, by Exercise 3.5.E. Section 57, while $V(g) \subset V(f) \;\Rightarrow\; D(f) \subset D(g)$, then g is invertible in A_f. Hence

$$\alpha(S_{D(f)}) = \{\alpha(g) \;\mid\; g \in S_{D(f)}\} \subset (A_f)^\times$$

and by universal property we can factor uniquely through A_f to $\mathscr{O}_{\operatorname{Spec} A}(D(f))$. Their composition must be identity hence we're done.

61.1 References

See [**?**] (2.10) on Page 59 for details.

62 4.1.B.

See next section Section 63

63 4.1.C.

See KEY LEMMA 1.13. on Page 5 of a book HERE.

Previously we proved that for arbitrary i, j

$$g_j^M \cdot b_i' = g_i^M \cdot b_j'. \star$$

In this lemma, the final equality is given by the following: let $f^k = \sum_{i \in I} a_i' g_i^M$ for some index set I and denote $b = \sum_{i \in I} a_j' b_j'$. For a fixed $i \in I$ with $j \in J$ arbitrary, we have

$$\begin{aligned} g_i^M b &= g_i^M \left(\sum_{j \in I} a_j' b_j' \right) \\ &= \left(\sum_{j \in I} a_j' g_i^M b_j' \right) \\ &= \left(\sum_{j \in I} a_j' g_j^M b_i' \right) \quad \text{by applying } \star \\ &= \left(\sum_{j \in I} a_j' g_j^M \right) b_i' = f^k b_i'. \end{aligned}$$

64 4.1.D.

64.1 Hint

See Def 01HT of [**?**], in which a lemma 00CR build connection between localisation and colimit. For this lemma, see also "A Term"...

For relations between colimit and localisation, see this post.

65 4.1.E.

See Section 64

66 4.1.F.

(a) See Atiyah [**?**] Chapter 3 Local Property.

(b) By 2.4.A. and Section 65, we have sheaf $\widetilde{M}$. According to Lemma 01HV (2), we have an injection

$$\widetilde{M}(\mathcal{O}_{\operatorname{Spec} A}) = M \to \prod_{[\mathfrak{p}] \in \operatorname{Spec} A} \widetilde{M}_{[\mathfrak{p}]} = \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}.$$

67 4.3.A.

See Hartshorne [**?**]?

68 4.3.B.

See Hartshorne[**?**] Exercise in Section 2.2.

69 4.3.C.

70 4.3.D.

By definition of a scheme, we know the union of all affine open subschemes will cover the scheme.

Given two affine open $(X_1, \mathcal{O}_{X_1}), (X_2, ...)$. For any point $p \in X_1 \cap X_2$, we know there exists an open neighborhood U such that $p \in U \subset X_1 \cap X_2$ given both X_1, X_2 are open. While we know distinguish sets form a base, there exists some

$$p \in D(f) \subset U \subset X_1 \cap X_2.$$

And by 68 or **??** we know $D(f)$ is an affine open as expected.

Therefore we've checked that affine opens form a base for Zariski topology. ?

71 4.3.E.

The topology on $\amalg X_i$ is box topology or product topology? I guess product topology... See a post regrading box topology and product topology.

71.1 (a)

Assume I is a finite index set. For each $i \in I$, we have an affine scheme $(X_i, \mathcal{O}_{X_i})$ where $X_i = \operatorname{Spec} A_i$ for some ring A_i. According to Exercise 3.6.A, we denote $A = \prod_{i \in I} A_i$. While we proved in [**?**] Chap II Exercise 1.9. **??**. that finite number of direct sum of sheaves is again a sheaf. We claim that

$$(\amalg_{i \in I} X_i, \oplus_{i \in I} \mathcal{O}_{X_i}) \simeq (\operatorname{Spec}(A), \mathcal{O}_{\operatorname{Spec} A}).$$

Exercise 3.6.A. checked two topological spaces are homeomorphic via $\pi : \amalg_{i \in I} X_i \to \operatorname{Spec} A$. It remains to check isomorphism of sheaves

$$\mathcal{O}_{\operatorname{Spec} A} \to \pi_*(\oplus_{i \in I} \mathcal{O}_{X_i}).$$

The map on sheaves is given by free if we assume Chap II Proposition 2.3. of [**?**] and its counterpart statement (which is slightly stronger) in here 67. Basic idea is that we have homeomorphism of two topological spaces, then apply the theorem, which will give us a unique morphism of sheaves. Do this conversely...

71.2 (b)

Suppose index set I is infinite. Assume for the sake of contradiction that $\amalg_{i\in I} X_i \simeq \operatorname{Spec} A$ for some ring A. Quasi-compact is preserved by homeomorphism, the $\operatorname{Spec} A$ is quasi-compact, but I claim the left hand side is not quasi-compact. ?

See 3.6.6.

71.3 References

See post regarding some concrete counterexamples.

Verification for part (a), post.

For non-compactness of (b). See a solution HERE by Pink. See post, then we know the space in (b) is not quasi-compact because we can choose a covering in which each open subset $U_i = A \times \cdot \times B_i \times \cdot A$ where $B_i \subsetneq A_i$. It cannot reduce to a finite sub-cover.

See Tag 00ED, Lemma 01I5 for more rigorous approach to (a).

72 4.3.F.

72.1 (a)

73 4.4.A.

Suppose we're given schemes $(X_i, \mathcal{O}_{X_i})$ indexed by $i \in I$.

Then we define

$$X = \amalg_{i\in I} X_i / \sim_{ij}$$

where $\sim_{ij}$ is generated by all f_{ij}. The structure sheaf is defined by glueing sheaves.

73.1 References

From a special case of S-scheme, [**?**] Lemma 3.33 of Chapter 2 on Page 49.

From Section 01JA of [**?**] on Glueing Schemes.

73.2 Cocycle Condition

Given Cocycle Condition, we claim that both "inverse" and "identity" requirements are redundent.

- For "identity": let all index to be i
- For "inverse": See the diagram (2) of Tag 01JA, we simply let $k = i$, which will give us $\varphi_{ji} \circ \varphi_{ij} = \mathrm{id}_{U_{ij}}$. Swap i, j will give us another equality, hence we must have $\varphi_{ij} = \varphi ji^{-1}$.

For confirmation, see [**?**] Definition 3.9. on Page 71.

74 5.1.C.

The hint in the prompt was refering to Exercise 2.13. **??** Chapter 2 of Hartshorne [**?**].

75 5.1.D.

Remarks: $\Rightarrow$ is easy because affine opens form a basis, while the converse direction is non-trivial. The point is that we only know a specific choice of affine opens.

Fact: we can build a larger quasi-compact topological spaces by union finitely many quasi-compact topological spaces. See this POST.

76 5.1.F.

$\Rightarrow$: Let U, V be two affine opens in scheme X. We know U, V are quasi-compact, then their intersection $U \cap V$ is quasi-compact. While affine opens form a basis for topology of X, by quasi-compactness of $U \cap V$ we can cover it by finitely many affine opens.

$\Leftarrow$: Let A, B be two quasi-compact open subsets of X. By quasi-compactness we can decompose them as

$$A = \bigcup_{i=1}^{n} A_i, \ B = \bigcup_{j=1}^{m} B_j.$$

where all A_i, B_j are affine opens. Notice that

$$A \cap B = \bigcup_{1 \leq i \leq n, \ 1 \leq j \leq m} (A_i \cap B_j)$$

is a finite union of $A_i \cap B_j$, by assumption it's a finite union of affine opens. Apply Section 75 we conclude $A \cap B$ is quasi-compact, which proves X is quasi-separated.

77 5.1.G.

Zariski topology admits distinguished open as a basis. For solution see this POST.

78 5.1.H.

$\Rightarrow$: If we take Section 76 as an equivalent definition of quasi-separated, then we can take the open affine cover as the original affine opens.

$\Leftarrow$: ???

78.1 Hint

See POST discussing this exercise.

79 5.1.K.

For details, see Tag 01KH.

80 5.2.A.

Proof.

See 3.2.1. When we say two functions f, g agree at all points, this means

$$f \equiv g \pmod{p}$$

for all $p \in \Gamma(X, \mathcal{O}_X)$.

To prove $f = g$, we wish to use the Identity Axiom of Sheaf. Fix an open covering $\{U_i\}_{i \in I}$ of X where I an index set. It suffices to check $f\mid_{U_i} = g\mid_{U_i}$.

We can identify them by injective map

$$\mathcal{O}_X(U_i) \hookrightarrow \prod_{p \in U_i} \mathcal{O}_{X,p}$$

This forces $f = g$ when they have same ???

□

80.1 References

See this POST: but is the definition of value of function correct?

See another POST.

81 5.2.C.

Localisation of reduced ring is again reduced for reduce is a local property.

For $\mathbb{A}_k^n$: polynomial ring $k[x_1, ..., x_n]$ is reduced.

For $\mathbb{P}_k^n$: it's similar, since taking subring will preserve reducedness.

82 5.3.1. Proposition

83

We'll try to interpret the equality in the proof by applying Hartshorne [**?**] Chap II Exercise 2.16.

$$\begin{aligned} D_B(g) &= \operatorname{Spec} B_g \\ &= \operatorname{Spec} B_g \cap \operatorname{Spec} A_f \\ &= (\operatorname{Spec} B)_g \cap \operatorname{Spec} A_f \quad \text{by Hart Ex 2.16} \\ &= D_{A_f}(g') \quad \text{by Hart Ex 2.16} \\ &= (\operatorname{Spec} A_f)_{g'} \\ &= \operatorname{Spec} A_{fg''}. \end{aligned}$$

Here g'' is specified as in the textbook [**?**], and g' is defined as

$$g \in B_g \to g' \in \Gamma(\operatorname{Spec} A_f, \mathcal{O}_X|_{\operatorname{Spec} A_f}) = A_f.$$

For $\operatorname{Spec} B_g = \operatorname{Spec}(A_f)_{g'}$, we need [**?**] Hartshorne Chap II Exercise 2.16.

83.1

Or we could do this...

Consider the following commutative diagram with corresponding ideals and elementes as labeled.

$$\begin{tikzcd} & g \in \mathfrak{p}' \triangleleft B \arrow[dl] \arrow[dr] & \\ g' \in \mathfrak{q} \triangleleft A_f \arrow[rr] & & g/1 \in \mathfrak{p} \triangleleft B_g \end{tikzcd}$$

By commutativity of the diagram, given a prime ideal $\mathfrak{p}$ then it corresponding uniquely to $\mathfrak{p}'$ and $\mathfrak{q}$; given $g/1$, then there exist g and g'.

$$\begin{aligned} \operatorname{Spec} B_g &\cong \{\mathfrak{p}' \triangleleft B \mid g \notin \mathfrak{p}'\} \\ &\cong \{\mathfrak{q} \triangleleft A_f \mid g' \notin \mathfrak{q}\} \\ &\cong \operatorname{Spec} A_f \setminus \{[\mathfrak{p}] \ : \ g' \in \mathfrak{p}\} = \operatorname{Spec}(A_f)_{g'}. \end{aligned}$$

84

Let X be a scheme with ***affine*** *open subsets $V \subset U \subset X$. Given $f \in \Gamma(U, \mathcal{O}_X)$ such that $\mathbf{D}_U(f) \subset V \subset U \subset X$. Then $\mathbf{D}_U(f) = \mathbf{D}_V(f|_V)$.* ?

85 5.3.2. Affine Communication Lemma.

See this POST.
Question: why stalk-local implies affine-local?

86 5.3.A.

Show that locally Noetherian schemes are quasi-separated.

Proof.
According to Section 78, it suffices to show the existence of a cover of X by affine open satisfies certain property. □

87 6.1.2. Theorem.

For the claim $\widetilde{M}|_{\mathsf{Spec}\, A_f} = \widetilde{M_f}$ See Theorem 1.7. Chap 5 on Page 160 [**?**].
The isomorphism of two sheaves of modules

$$\widetilde{M_i}|_{\mathbf{D}(f_i)\cap\mathbf{D}(f_j)} \simeq \widetilde{M_j}|_{\mathbf{D}(f_i)\cap\mathbf{D}(f_j)}$$

yield isomorphisms $\phi_{ij} : (M_i)_{f_j} \simeq (M_j)_{f_i}$ of $A_{f_i f_j}$-modules. Notice that M_i admits A_{f_i}-module structure as assumed. Therefore,

$$\widetilde{M_i}|_{\mathbf{D}(f_i)\cap\mathbf{D}(f_j)\subset\mathsf{Spec}\, A} = \widetilde{M_i}(\mathbf{D}_{\mathsf{Spec}\, A_{f_i}}(\overline{f_j})) = (M_j)_{f_j}$$

where $\overline{f_i}$ denotes the image of f_i under restriction from $\Gamma(\mathsf{Spec}\, A_{f_i}, \mathcal{O}_{\dots})$ to $\Gamma(\mathsf{Spec}\, A_{f_i}|_{\mathsf{Spec}\, A_{f_j}, \mathcal{O}_{\dots}})$. Here the second equality applied result of Ex 2.16. Chap 2 [**?**].

Todo List

1. Define β
2. Check injective, and $\mathsf{Im}\,\beta = \mathsf{Ker}\,\gamma_{f_1}$.
3. Apply five lemma to conclude the isomorphism between $M_{f_i} \simeq M_1$.

Then the induced map in the category of quasi-coherent sheaves will be an isomorphism as expected.

To define β, it suffices to define $M_1 \to (M_i)_{f_1}$.

$$\widetilde{M_1} = \mathcal{F}|_{\mathbf{D}(f_i)} = \mathcal{F}|_{D(f_i)\cap D(f_1)} = \dots = \widetilde{M_i}|_{\mathsf{Spec}\, A_{f_1}}.$$

???

87.1 Remarks

The part for verifying the first condition of Affine Communication Lemma is clear. We'll focus on checking the second condition. Here we're given information for $\mathsf{Spec}\, A_{f_i}$ for $1 \le i \le n$. By definition of property P, we have $\mathscr{F}$ is $\mathscr{O}_{\mathsf{Spec}\, A}$-module and $\mathscr{F}|_{D_{\mathsf{Spec}\, A}(f_i)} = \widetilde{M_i}$. We wish to prove $\mathsf{Spec}\, A$ has the property P.

?? but as the statement of affine communication lemma, the information we get is $\mathsf{Spec}\, A_{f_i} \to X$, but even in that case, we just restrict and regard $\mathscr{F}$ as a sheaf on $\mathsf{Spec}\, A$

The text indicated that

$$\mathscr{F}|_{\mathbf{D}(f_i)\cap\mathbf{D}(f_j)} \simeq \widetilde{M_i}|_{\mathbf{D}(f_i)\cap\mathbf{D}(f_j)}.$$

More precisely, this could be written as

$$\begin{aligned} \phi_i : \mathscr{F}|_{\mathbf{D}_{\mathsf{Spec}\, A}(f_i)} &\simeq \widetilde{M_i} \\ \Rightarrow\ \mathscr{F}|_{\mathbf{D}_{\mathsf{Spec}\, A}(f_i)\cap\mathbf{D}_{\mathsf{Spec}\, A}(f_j)} &\simeq \widetilde{M_i}|_{\mathbf{D}_{\mathsf{Spec}\, A}(f_i)\cap\mathbf{D}_{\mathsf{Spec}\, A}(f_j)} \\ \Rightarrow\ (F|_{\mathbf{D}_{\mathsf{Spec}\, A} f_i})|_{\mathbf{D}_{\mathsf{Spec}\, A_{f_i}}(\overline{f_j})} &\simeq \widetilde{M_i}|_{\mathbf{D}_{\mathsf{Spec}\, A_{f_i}}(\overline{f_j})}. \end{aligned}$$

Here we denote $\overline{f_j} = (\mathsf{Res}_{..} : \mathsf{Spec}\, A \to \mathsf{Spec}\, A_{f_i})(f_j)$, i.e. we denote $\overline{f_j}$ as the image of f_j under restrction to $\mathsf{Spec}\, A_{f_i}$. Notice that $\widetilde{M_i}$ is a sheaf on $\mathsf{Spec}\, A_{f_i}$, which gives us

$$\widetilde{M_i}|_{\mathbf{D}_{\mathsf{Spec}\, A_{f_i}}(\overline{f_j})} = (M_i)_{\overline{f_j}}.$$

?

So a more precise way to describe ϕ_{ij} is by indicating that the function f_j is the image $\overline{f_j}$, or

$$\phi_{ij} : (M_i)_{\overline{f_j}} \to (M_j)_{\overline{f_i}}.$$

In textbook it was regarded as $A_{f_i f_j}$-module. More precisely, it localised twice.

The assumption of condition 2 give us existence of $\mathscr{F}$ (on $\mathsf{Spec}\, A$, if not then restrict), we need to provide the existence of A-module M such that $\widetilde{M} \simeq \mathscr{F}$. We define it as the kernel of the exact sequence as indicated in the textbook.

What we need to do is to glue sheaves from each M_i to M. In other words, we're trying to check isomorphism of sheaves on distinguished opens.

$$\forall\, \mathsf{Spec}\, A_{f_i}, \widetilde{M}|_{\mathsf{Spec}\, A_{f_i}} = \mathscr{F}|_{\mathsf{Spec}\, A_{f_i}} = \widetilde{M_i} \ \Leftrightarrow\ \widetilde{M} = \mathscr{F}.$$

To define β, we recall that we've defined ϕ_{ij}, then we can use them to define

$$\begin{aligned} \beta : M_1 &\to (M_1)_{f_1} \times \cdots \times (M_n)_{f_1} \\ m &\mapsto ((m/1), \phi_{12}((M_1 \to (M_1)_{f_2})), ..., \phi_{1n}((M_1 \to (M_1)_{f_n})) \end{aligned}$$

$$\begin{tikzcd} & M_1 \arrow[dl] \arrow[dr, dashed] & \\ (M_1)_{f_2} \arrow[rr, "\phi_{12}"] & & (M_2)_{f_1} \end{tikzcd}$$

Clearly this map is injective. Remains to check $\mathsf{Im}\, \beta = \mathsf{Ker}\, \gamma_{f_1}$.

How?

Or another approach is to identify (6.1.2.3) and (6.1.2.4) term by term. The later (6.1.2.4) is exact and β will be obvious to define.

87.2 References

See NOTES. Or Proposition-Definition 5.1. of Algebraic Geometry II on Page 20 [**?**].

88 6.1.B.

Here is a POST discussing a non-example of quasi-coherent sheaf. And HERE.

See this conversation HERE and HERE.

89 6.2.1. Definition: Distinguished Affine Base

See this POST.

90 6.2.D.

$\Rightarrow$: Suppose $\mathscr{O}_X$ module $\mathscr{F}$ is quasi-coherent, then the above diagram gives us $\Gamma(\mathsf{Spec}\, A, \mathscr{F}) = M$ for some A-module M.

$\Leftarrow$:

91 6.2.G.

92 6.4.A.

92.1 (i)

Localisation (with respect to f) is exact. Hence the following is exact

$$(R^{\oplus n})_f = (R_f)^{\oplus n} \to M_f \to 0.$$

Another way to interpret this is by noticing that the generator $m_1, ...m_n \in M$ will be send to generators of M_f as A_f-module.

92.2 (ii)

We can define $\phi : M \to \prod_{1 \leq i \leq n} M_{f_i}$ by $m \mapsto (m/1, m/1, ..., m/1)$, i.e. use universal property of product with natural map of localisation.

Now we have M_{f_i} is f.g. as A_{f_i}-module hence we can assume there's a surjective for each i

$$(A_{f_i})^{\oplus l_i} \to M_{f_i} \to 0$$

and we denote $L = \sum_{1 \leq i \leq n} l_i$.

$$\begin{tikzcd} A^{\oplus L} \arrow[r, dashed] \arrow[d, hook, "\psi"'] & M \arrow[d, hook, "\phi"] \\ \prod_{1\le i\le n} A_{f_i}^{\oplus l_i} \arrow[r, two heads, "\prod \gamma_i"'] & \prod_{1\le i\le n} M_{f_i} \end{tikzcd}$$

Now we wish to check ϕ is injective. It suffices to check $\mathsf{Ker}\,\phi \subset 0$. For any $m \in M$ such that $\phi(m) = (m/1, ..., m/1) = 0$, we wish to show $m = 0$. Notice that $m/1 = 0 \in M_{f_i}$ implies $f_i^{k_i} m = 0 \in M$ for some integer k_i. Therefore we can pick $K = \sum k_i$ (or other finite integer large enough)

$$m = 1m = (a_1 f_1 + ... + a_n f_n)m = (a_1 f_1 + ... + a_n f_n)^K m = 0.$$

For any $m \in M$, we identify it with

$$\phi(m) = (\pi_1 \circ \phi(m), ..., \pi_n \circ \phi(m)) \in \prod_{1\le i\le n} M_{f_i}$$

where π_i be projection to i-th component. Then $\pi_i \circ \phi(m) \in M_{f_i}$, and under each surjection γ_i it will correspond to finitely many generators $b_{i,j}/f_i^\bullet \in M_{f_i}$ where $1 \le j \le l_i$. For each $1 \le i \le n$, we know

$$\begin{aligned} M_{f_i} &= A_{f_i} \langle b_{i,1}/f_i^\bullet, b_{i,2}/f_i^\bullet, ..., b_{i,l_i}/f_i^\bullet \rangle \\ &= A_{f_i} \langle b_{i,1}, b_{i,2}, ..., b_{i,l_i} \rangle \end{aligned}$$

Or we can say there's a surjective map from $A^{\oplus l_i} \to M_{f_i}$, and we can inductively build a surjective map from $A^{\oplus L} \to \prod_{1\le i\le n} M_{f_i}$. Restrict the codomain of this map will give us a surjection that proves M is finitely generated A-module

$$A^{\oplus L} \to M.$$

92.3 Comments

Localisation commute with arbitrary direct sum HERE. See Tag 01OR, and Zariski-local properties of modules and algebras Tag 00EO. See this POST. For official reference, see Tag 00EN. Here the actual proof relies on Lemma 1 of Tag 00EO.

Lemma: If each $M_{f_i} = 0$, then $M = 0$.

Proof.
For $\mathfrak{p} \in \mathsf{Spec}\, A$, we can cover it by some $\mathbf{D}(f_i)$. In particular, $f_i \notin \mathfrak{p} \lhd A$.

A result about further localisation Tag 02C6 will help us as

$$\begin{aligned} (M_{f_i})_{\mathfrak{p}} &= \overline{\{A - \mathfrak{p}\}^{-1}}(\{1, f_i, f_i^2, ...\}^{-1} M) \\ &= (\{A - \mathfrak{p}\}\{1, f_i, f_i^2, ...\})^{-1} M \\ &= (\{A - \mathfrak{p}\})^{-1} M = M_{\mathfrak{p}}. \end{aligned}$$

Given the assumption that $M_{f_i} = 0$, then $M_{\mathfrak{p}} = 0$ for arbitrary $\mathfrak{p} \in \mathsf{Spec}\, A$. And now we're ready to apply "being 0 is a local property" Tag 00HN to finish the proposition.

□

93 6.4.6. Lemma:

See this POST with an elegant solution. Note $\tilde{f}$ is defined by simply restriction. In this post, it's worth mention Tag 07JX.

See Ex. 5.18. [**?**]. This approach requires Schanuel's Lemma 5.17. [**?**].

94 6.4.B.

95 6.4.C.

Similary to Section 94, for references see Tag 00EN.

96 6.6.B.

For the counterexample, see HERE and HERE for a concrete description of computing localisation of quotient rings in a simple setting. And in general we have this example HERE.

Support of a module need not be closed in general.

Proof.
According to the hint, we take $A = \mathbb{Z}$ and

$$M = \oplus_{p \text{ prime}} \mathbb{Z}/p\mathbb{Z}.$$

Notice that in Principal Ideal Domain $\mathbb{Z}$, we have for two primes $p, q \in \mathbb{Z}$,

$$(\mathbb{Z}/p\mathbb{Z})_{(q)} = \begin{cases} 1 \\ \mathbb{Z}/p\mathbb{Z} \end{cases}$$

Because when $p \neq q$, then...

□

For a more general treatment, see Example 6.25. of Gathmann's Notes in Commutative Algebra P58. And HERE.

97 6.6.I.

Proof.
Kernel K is a nonzero submodule of M over a Noetherian ring A. Therefore it admits an associate prime $\mathfrak{p} \triangleleft A$ that is the annihilator of some element $m \in K \subset M$. Notice the injection

$$A/\mathfrak{p} \to K \to M,$$

then $\mathfrak{p} \in \mathsf{Ass}\, M$. Then under the natural map,

$$m \mapsto m/1 \in M_{\mathfrak{p}}$$

is nonzero. □

98 6.6.16.

As $\mathbb{Z}$-module $\mathbb{Z}$ contains 0 as zero-divisor. Therefore

$$\mathsf{Ass}_{\mathbb{Z}}\,\mathbb{Z} = \{0\}.$$

On the other hand, we have

$$\mathsf{Ass}_{\mathbb{Z}}\,\mathbb{Z} \cup \mathsf{Ass}_{\mathbb{Z}}\,\mathbb{Z}/2\mathbb{Z} \subset \{0, (2)\}.$$

99 6.6.J.

See Lemma (17.16). [**?**]. Such a cute proof...

100 6.6.L.

Recall that the union of all associate primes consisits only zerodivisors. While $A = k[x, y]$ is an integral domain, any filtration must contains some nonzero primes, which cannot be associate prime.

101 7.2.A.

MORPHISMS OF RINGED SPACES GLUE.

Proof.
For topological map f, the existence is clear (existence and the restriction will behave as desired). It remains to define the morphism of sheaves

$$f^\sharp : \mathscr{O}_Y \to \pi_* \mathscr{O}_X$$

and check that $(\pi|_{U_i})^\sharp : \mathscr{O}_Y \to (\pi|_{U_i})_* \mathscr{O}_{U_i} = \pi_i^\sharp : \mathscr{O}_Y \to \pi_{i,*}\mathscr{O}_{U_i}$.

By assumption, we have $\{\pi_i \,:\, U_i \to Y\}_{i\in\bullet}$ with $\pi_i^\sharp : \mathscr{O}_Y \to \pi_{i,*}\mathscr{O}_{U_i}$ for each $i \in \bullet$. Also recall that

$$\begin{aligned}(\pi_i|_{U_i\cap U_j})^\sharp : \mathscr{O}_Y \to (\pi_i|_{U_i\cap U_j})_* \mathscr{O}_{U_i\cap U_j} = \\ (\pi_j|_{U_i\cap U_j})^\sharp : \mathscr{O}_Y \to (\pi_j|_{U_i\cap U_j})_* \mathscr{O}_{U_i\cap U_j}.\end{aligned}$$

Therefore for any open subset $W \subset Y$, we have $(\pi_i|_{U_i\cap U_j})^\sharp(W) = (\pi_j|_{U_i\cap U_j})^\sharp(W)$. In particularly, we have

$$\begin{aligned}(\pi_i|_{U_i\cap U_j})_* \mathscr{O}_{U_i\cap U_j}(W) &= \mathscr{O}_{U_i\cap U_j}((\pi_i|_{U_i\cap U_j})^{-1}(W)) \\ &= \mathscr{O}_{U_i\cap U_j}(\pi_i^{-1}(W) \cap U_i \cap U_j) \\ &= \mathscr{O}_X(\pi_i^{-1}(W) \cap U_i \cap U_j) \\ &= \mathscr{O}_X(\pi_j^{-1}(W) \cap U_i \cap U_j) \\ &= (\pi_j|_{U_i\cap U_j})_* \mathscr{O}_{U_i\cap U_j}(W).\end{aligned}$$

Now we need to define $\pi^\sharp : \mathscr{O}_Y \to \pi_* \mathscr{O}_X$. For every open subset $W \subset Y$, we have to define a map

$$\pi^\sharp(W) : \mathscr{O}_Y(W) \to \pi_* \mathscr{O}_X(W).$$

For every $s \in \mathscr{O}_Y(W)$, we define(at this moment, we've constructed topologically the map π)

$$\begin{aligned} s_i := \pi_i^\sharp(W)(s) &= (\mathscr{O}_Y(W) \to \pi_{i,*}\mathscr{O}_{U_i}(W))(s) \\ &\in \mathscr{O}_X(\pi_i^{-1}(W) \cap U_i) = \mathscr{O}_X(\pi^{-1}(W) \cap U_i) \end{aligned}$$

We can check

$$\begin{aligned} s_i|_{\pi^{-1}(W)\cap U_i \cap \pi^{-1}(W)\cap U_j} &= (\mathscr{O}_X(\pi^{-1}(W) \cap U_i) \to \mathscr{O}_X(\pi^{-1}(W) \cap U_i \cap U_j))(s_i) \\ &= (\mathscr{O}_X(\pi_i^{-1}(W) \cap U_i) \to \mathscr{O}_X(\pi_i^{-1}(W) \cap U_i \cap U_j))(s_i) \\ &= (\mathscr{O}_Y(W) \to \mathscr{O}_X(\pi^{-1}(W)) \to \mathscr{O}_X(...))(s) \\ &= (\mathscr{O}_X(\pi_j^{-1}(W) \cap U_j) \to \mathscr{O}_X(\pi_j^{-1}(W) \cap U_i \cap U_j))(s_j) \\ &= s_j|_{\pi^{-1}(W)\cap U_i \cap \pi^{-1}(W)\cap U_j} \end{aligned}$$

Notice that we have $\{\pi^{-1}(W) \cap U_i\}_{i\in\bullet}$ covers $\pi^{-1}(W)$ and $\mathscr{O}_X$ is a sheaf. Therefore there exist $S \in \mathscr{O}_X(\pi^{-1}(W)) = \pi_*\mathscr{O}_X(W)$ such that $S|_{U_i} = s_i$. Hence for every $W \subset Y$ we've defined the map $\pi^\sharp$. And we can check that for all open subsets $W \subset U_i$,

$$(\pi|_{U_i})^\sharp(W) = \pi_i^\sharp(W) \Rightarrow (\pi|_{U_i})^\sharp = \pi_i^\sharp.$$

$\square$???

101.1 Comments

See Prop 3.5. [**?**] on Page 70. For details, see "Glueing morphisms of sheaves together - can I just do this?" and "morphism of ringed spaces glue".

102 7.2.E.

See Hartshorne [**?**] Page 72 for details.

103 7.3.A.

For a local homomorphism, we have

$$\mathfrak{m}_q \subset \mathscr{O}_{Y,q} \to \mathscr{O}_{X,p} \to \kappa(p).$$

It will induce a map $\kappa(q) \to \kappa(p)$.

104 7.3.B.

105 7.3.D.

!

106 7.3.G.

See [**?**] Proposition 3.4. Chap 3 on Page 69.
See Tag 01I1.

107 7.3.6.

For "fun question": is that maximalSpec?

108 7.3.L.

Nice hint. See Hartshorne Chap 2 Exercise 2.7. [**?**].

109 7.4.

Skipped...

110 7.5.B.

Tag 0CC1

111 7.5.4.

Gathmann's notes Definition 9.5 provided the proof, with one subtlety missed. See Gemini: Dominant Maps and Rational Map Composition

112 8.1.1.

112.1 Remarks

See a post discussing "Does "local on the target" mean the same thing as "local on the base"?" HERE.

113 8.1.A.

Proof.
Notice that $X \times_S Y' \simeq X \times_Y Y \times_S Y'$ and $X' \times_{Y'} X \times_S Y' \simeq X' \times_{Y'} Y' \times_S X \simeq X \times_S X'$. Therefore apply (ii) and (i) we can conclude $X \times_S X' \to Y \times_S Y'$ has property P. □

114 8.1.B.

Proof.
Requirement (i) is clear.

For (ii), we have the following diagram

$$\begin{tikzcd} X \times_Y Y' \arrow[r] \arrow[d] & Y' \arrow[d] \\ X \arrow[r, "\simeq"] & Y \end{tikzcd}$$

Observe that $X \times_Y Y' \simeq Y \times_Y Y' \simeq Y'$. Hence (ii) is satisfied.

For (iii), part (a) is clear. It remains to consider part (b). And for this part we refer to [**?**] Exercise 2.17. (a) on Page 81 of Chapter 2. □

115 8.2.B.

115.1 (a)

See this post HERE on StackExchange. To show a ring morphism is not integral, we can consider the lying over property.

116 8.3.B.

116.1 (a)

See 5.3.4. [**?**].

Proof.
Notice that any morphism of schemes consists of a continuous map such that

$$\pi^{-1}(U) \subset X$$

is open. Then we apply Exercise 5.1.C., which shows that X is a Noetherian topological space. Apply Exercise 3.6.U. shows that any open subset, including $\pi^{-1}(U)$ is quasi-compact. And this proves that this morphism π is quasi-compact. □

116.2 (b)

?

117 8.3.D.

Clear. Because affine scheme is both quasi-compact and quasi-separated.

118 8.3.F.

For example (3), see Exercise 9.12. by Gathmann notes on Commutative algebra [**?**], Page 83.

119 8.3.G.

Proof.
Since finite morphism is by definition affine, it suffices to show the statement for $X = \mathsf{Spec}\, A$ an affine scheme. While finite morhism is integral, then we know the inclusion $k \hookrightarrow A$ is finite with A being an integral domain. Apply Exericse 3.2.G, we conclude A is a field.

□

120 8.3.H.

Apply 8.2.C. and use the definition "for all".

121 8.4.A.

failed...
$\Rightarrow$ is immediate. $\Leftarrow$: in general, we can express constructible subset A as a finite intersection of family members

$$A = \bigcap_{\leq \infty} A_i$$

where each $A_i \in \mathscr{A}$ is a constructible subset. □

*sketch, Proposition 10.13. [**?**].*
So the problem here is to prove two equivalent conditions:

1. smallest subsets satisfying three conditions;
2. finite union of locally closed subsets;
3. finite disjoint union of locally closed subsets.

We'll use the very fact "smallest". Our plan is to check that all subsets of (2) satisfy the three conditions in (1): this is immediately clear for it contains all open subsets and are closed under taking finite unions. It remains to check it's closed under taking complement... □

122 8.4.B.

Remark 1. *My idea was to find a map $X \to \mathbb{A}^1$ such that $0 \subset X$ is clearly not constructible. But I failed. For hint, see HERE. Be careful for some details.*

Proof.
Suppose $\{\eta\}$ is constructible, i.e. we have $\eta = \bigcup_{\leq \infty} U_i \cap V_i$ where each $U_i \cap V_i$ are disjoint. Then we focus on one pair for $\{\eta\}$ is a singleton: let $0 = \{\eta\} = U \cap V$ where U is open and V is closed. While η is the generic point, then $\eta \mathbb{A}^1$. Therefore $\{\eta\} = U$ as a finite subset of $\mathbb{A}^1$ is open. This contradicts the fact that all closed subset of $\mathbb{A}^1$ are $\emptyset$, finite subsets, and all of the space $\mathbb{A}^1$. $\square$

123 11.1.A.

According to the diagram, notice that

$$X \times_Y Y \times_Y Y \times_Z Y \simeq X \times_Z Y,$$
$$X \times_Z Y \times_X Z \simeq Z \times_Z Y.$$

124 11.1.E. Exercise

First of all, structure morphism must automatically lies in P because P contains all isomorphisms, therefore applying $S \to S$ base change gives us $S \times_S A = A \to S$ has property P.

$$\begin{tikzcd} X \arrow[rr, "\therefore \in P"] \arrow[dr, "\in P"'] & & Y \times_S Z \arrow[dl, "\in P\delta"] \\ & S & \end{tikzcd}$$

125 11.2.C.

Use Exercise 1.2.W, we have the diagonal morphism is an isomorphism. Hence closed embedding.

126 11.2.E.

Assume X is separated over $\mathsf{Spec}\, A$, $\mathsf{Spec}\, A \to \mathsf{Spec}\, \mathbb{Z}$ is an affine morphism, we can compose two separated morphism to conclude X is separated (over $\mathsf{Spec}\, \mathbb{Z}$).

Conversely, consider the following diagram:

$$\begin{tikzcd} X \arrow[rr, "\therefore P"] \arrow[dr, "\in P"'] & & \operatorname{Spec} A \arrow[dl, "\in P\delta"] \\ & \operatorname{Spec} \mathbb{Z} & \end{tikzcd}$$

It suffices to check $\mathsf{Spec}\, A \to \mathsf{Spec}\, A \times_{\mathsf{Spec}\, \mathbb{Z}} \mathsf{Spec}\, A$ is separated. But this is already an affine morphism so we're done.

127 11.2.F.

127.1 (a)

locally closed embedding.
For locally closed embedding, we apply Cancellation Theorem for property P of being a "local closed embedding". With the existing assumption, it suffices to check ρ is $P\delta$. Recall $Y \to Y \times_Z Y$ is a local closed embedding by 11.2.1. Proposition, hence we're done. □

locally of finite type.
It boils down to check ρ is $P\delta$ for property P of being a "locally of finite type" morphism. Based on previous part, we know $Y \to Y \times_Z Y$ is a local closed embedding. Hence it's locally of finite type by 9.2.A. □

separated.

□

Failed to give a proof. See Tag 01KV.

128 11.2.G.

11.2.F. with Cancellation Theorem for property P of either locally closed embedding or separated morphism can solve the first two parts.

Proof.

□

129 11.3.A.

Tag 01KM

130 11.3.C.

Let $\pi : W = \mathbb{A}^1 \to X$ to be the inclusion of one copy of affine line $\mathbb{A}^1$, and π' to be the inclusion from another copy. They agree on $\mathbb{A}^1 \setminus \{0\}$, which is NOT closed. According to 11.3.A., if X is separated, then two morphisms should agree on a closed subscheme, contradiction.

131 11.3.2.

See Tag 01RR and Math Stack Exchange discussion.

132 12.1.B.

Proof.

$$\left\{ \text{ irreducible closed subsets of } U \ \right\} \longrightarrow \left\{ \begin{array}{l} \text{irreducible closed subsets of } X \\ \text{that meet } U \end{array} \right\}$$

$$\begin{aligned} A &\mapsto \overline{A} \quad \text{(closure taken in } X) \\ A \cap U &\leftarrow\!\shortmid A \end{aligned}$$

□

133 12.1.J.

Read 12.1.4. second paragraph: "In particular, the codimension of a point ..."

134 12.1.K.

Concretely, two chains on the left hand side give rise to a chain of irreducible closed subsets in topological space X. While the definition $\dim X$ requires that the chain to be sup, thus we have the inequality $\leq$.

135 13.5.G.

See this POST on Math Stack Exchange.

But what is the precise definition of valuation in this context? See Tag 0BE0.

136 14.1.C.

Taking possible intersections of the covers for which both locally free sheaves are free, we may reduce the problem to check

$$(\mathcal{O}^{\otimes mn})|_{U_i} \cong \mathcal{H}om(\mathcal{F}, \mathcal{G})|_{U_i}.$$

And this is a direct consequence of the following observation: Page 179, equatioin (7.4.11) in [**?**].

137 14.1.D.

Proof.
In this proof, we'll utilise the following algebraic fact: For any ring R and any R-module M, we have an isomorphism

$$M \cong M^{\vee\vee}.$$

In fact, this solves the problem over affine schemes. Applying this fact repeatedly, we've defined the morphisms over an open cover of our domain. Thus we only need to check if the morphism glues together. Over the intersection, they agree because localisation commutes with dual functor (i.e. the hom functor). □

Proof.
In this attempt, we'll use transition maps to define our $\mathscr{E}^\vee$. Let $\{U_i\}$ be an open cover of X such that $\psi_i : \mathscr{E}|_{U_i} \cong \mathcal{O}_{U_i}^{\oplus n}$. Then our transition maps are given by

$$T_{ij} := \psi_i \circ \psi_j^{-1} \in \mathsf{GL}_n(\mathcal{O}_X(U_i \cap U_j)).$$

Now we need to examine the dual sheaf $\mathscr{E}^\vee$. Over U_i, we need a suitable choice for the isomorphism

$$\varphi_i : \mathscr{E}^\vee|_{U_i} \to \mathcal{O}_{U_i}^{\oplus n} \cong (\mathcal{O}_{U_i}^{\oplus n})^\vee. \tag{1}$$

And our choise will be $\varphi_i := \psi_i^{\vee,-1}$. Therefore our new transition maps for $\mathscr{E}^\vee$ are given

$$\begin{aligned} S_{ij} &= \varphi_i \circ \varphi_j^{-1} && (2)\\ &= (\psi_i^{\vee,-1}) \circ (\psi_j^{\vee,-1})^{-1} && (3)\\ &= (\psi_i^{\vee,-1}) \circ \psi_j^\vee && (4)\\ &= (\psi_j \circ \psi_i^{-1})^\vee && (5)\\ &= T_{ij}^{-1,\vee}. && (6) \end{aligned}$$

Thus applying twice the dual functor, the transition maps will restore the original T_{ij}, and we are done. □

138 14.2.A.

My discussion with ChatGPT HERE
See this POST for an easier solution.

139 18.5.A.

139.1 (a)

Assume (b).

$$\begin{aligned} h^0(C, \omega_C) &= h^0(C, \omega_C \otimes \omega_C^\vee) + 1 - g + \deg \omega_C \\ &= h^0(C, \mathcal{O}_C) + 1 - g + 2g - 2 \\ &= 1 + 1 - g + 2g - 2 = g. \end{aligned}$$

139.2 (b)

See Chap IV Example 1.3.3. [**?**].

140 23.1.C.

Almost by definition...

140.1 References

See this POST, and a detailed POST.
For details, see [**?**] Theorem 7.2. on Page 405.

141 23.4.F.

See remarks of **??**.